

Curriculum Vitae

Summary:

Final year MEng Mechanical Engineering student with industrial experience in the nuclear industry. Experienced in combining fundamental judgement with quantitative analysis to address technical challenges. Academic sponsor recipient with a proven ability to lead technical reporting and client-facing deliverables.

2021-2026: University of Southampton, MEng Mechanical Engineering with Placement Year at AtkinsRéalis.

- **MEng Group Project: 'Aerial Seeding Fixed-Wing UAV'.**
 - Structural design lead on a 5-person team delivering a 17 kg, 3.35 m wingspan fixed-wing UAV capable of autonomously dispersing seed bombs. Responsible for: regulatory load-case derivation (CS-23/FAR 23, 4.5g manoeuvre and gust); FEA (ANSYS Mechanical); fuselage CAD and structural layout; historic weather analysis (Python/MeteoStat); and composites analysis of CFRP spars (ESAComp, CLPT). Aircraft flew successfully with primary structure intact.
- **BEng Project** in High Temperature Superconductors: 'Investigating the effectiveness of a Low-Oxygen, Temperature Controlled Soldered Copper to REBCO Joint.'
- **Third Year Design Module:** Real-World modelling of cost savings from small-scale wind power (Python - Pandas/MeteoStat)
- **Proficient in:** Python for Data Science and Machine Learning (Jupyter, Numpy, Matplotlib, Pandas, Keras, Pytorch), Finite Element Analysis (ANSYS Mechanical), CAD Software, MATLAB, C++, Excel, COMSOL heat transfer modelling, ESAComp (composites laminate analysis).

Industrial Experience

Placement Year and Summer Placement, AtkinsRéalis Nuclear & Power: 2023-2025:

Industrial experience within Reactor Operations, working on projects to make quantitative and qualitative engineering judgements regarding the safety of operational, decommissioning and new build nuclear power stations. This involved writing technical reports, performing hand calculations/numerical analysis, interacting with clients and bid writing to win new work. Awarded academic sponsorship following placement.

Projects:

- **Meteorological Hazards Extent of Condition Report:**
 - Assisted in authorship of a fleetwide review of station resilience to meteorological hazards, addressing climate change impacts; recognised for high-quality contributions.
- **PWR Turbine Disintegration Licensing Summary Statement:**
 - Conducted Monte-Carlo analysis of steam turbine failure event using an in-house tool and authored an appendix summarising the results to contribute to licensing documentation for a £38bn new build project.
- **Moisture Monitoring Engineering Advice Note (EAN).**
 - Authored a report justifying safe relaxation of in-reactor moisture limits for a reactor in the defuelling stage for reduction in operator-hours and improved defuelling efficiency.
- **Dungeness B OEM Neutron Scatter Plug EAN:**
 - Lead author of an 8500-word report. Presented analysis of prior testing data giving confidence to understanding of metallurgical behaviours and likelihood of fault within nuclear fuel assembly.
- **Ad – Hoc Commercial and Technical work.**
 - Supported bid-writing and project management activities.
 - Developed a tool to calculate gas purging requirements for reactor and auxiliary equipment.

Transferable Experience

Renewable Energy and Sustainability Society, 2021-2023: Grew society profile and engagement as a founder.

Headlands Garage/Cylinder Head Services, 2018-2019: Supported automotive technicians; gained hands-on knowledge.

KLM Taxis Telephone Operator, 2019-2022: Handled bookings and built rapport with customers and drivers.

Interests and Other Skills

Cycling: Top 20 achieved in national time trial championship (2024). University 10-mile TT record holder and elite road cyclist with sponsored team DRAFT Racing. **Languages:** French (Basic/Intermediate), German (Intermediate).